

6 DNA and RNA are polynucleotides.

- (a)** Describe **three** ways in which the structure of messenger RNA (mRNA) differs from the structure of DNA.

In each of your answers, include information about the structure of mRNA **and** the structure of DNA.

1

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2

.....

3

.....

[3]

- (b)** Scientists have synthesised four synthetic bases, Z, P, S and B. The base pairings of the synthetic bases are shown in Fig. 6.1.

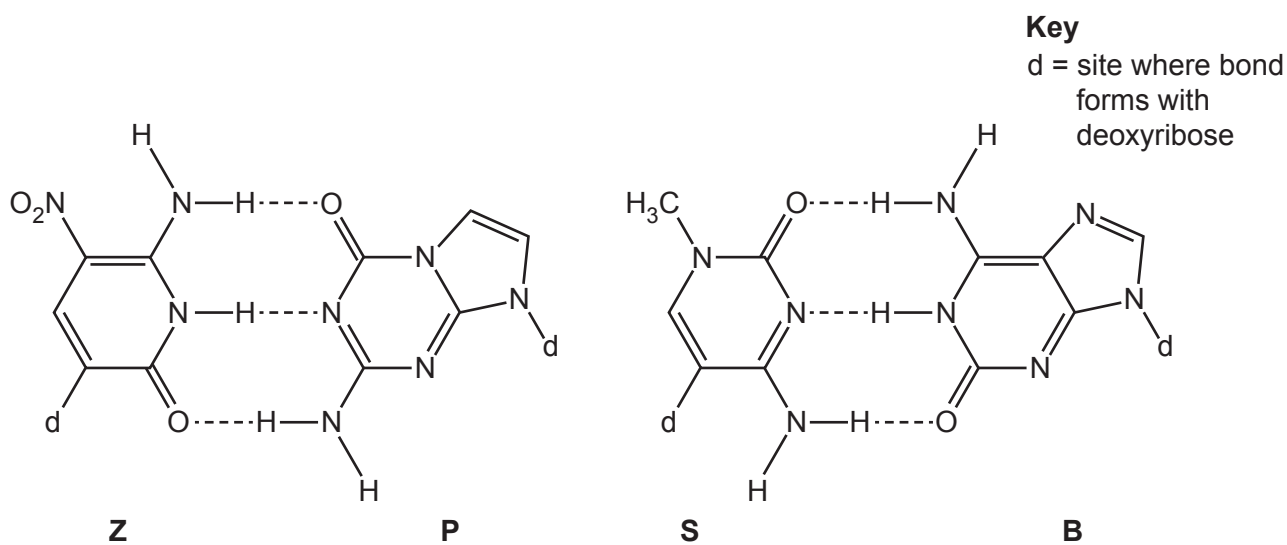


Fig. 6.1

(i) State the letters that represent the two purine synthetic bases in Fig. 6.1.

..... [1]

(ii) State **and** explain which DNA base pair is most similar to the synthetic base pairs in Fig. 6.1.

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..... [2]

[Total: 6]